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Causal Inference Homework 3

Part 1: Manipulation of Running Variable

1. Test for manipulation of the running variable (i.e. election rigging). One way to do this in Stata is to use rddensity. Explain what you are testing for and why.

The first part of the homework asks us to test manipulation of the running variable. In laymean's terms this means we want to test if the election vote margin is being artificially manipulated (such as rigging elections so that more close elections appear on the Democrats winning side than the losing – implying they may be cheating close elections). To test this hypothesis I use rddensity as the homework asks and get a p-value = 0.1226 and a robust p-value = 0.4579. These p-values tell us we cannot reject the null hypothesis that there is no manipulation of the running variable at the p=0.01, 0.05, 0.1 levels. I also made a historgram plot (Figure 1) showing the density for elections where the election was decided by 5% or less – an eye test confirms that there is no increase in density on right-hand side (blue).

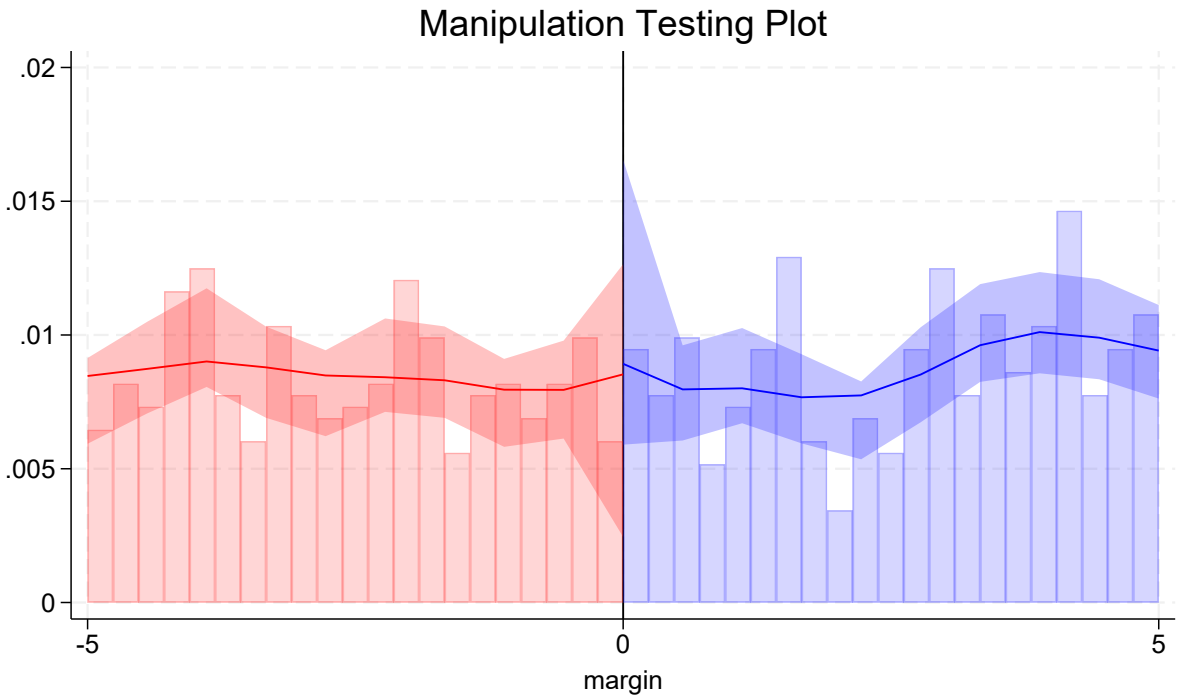

Figure 1: Manipulation of the Running Variable

Part 2: Treatment Effect – Continuity Assumption

2. Now let's estimate the treatment effect, using the continuity approach. Using rdrobust, first check the 5 covariates for continuity at the cutoff (zero). Explain why you are doing this and what you found. For the purposes of the homework, even if some of them flunk, we will proceed to estimate a treatment. Second, estimate the model using rdrobust. Let the software pick the bandwidth and polynomial. Report your results, plot them using rdplot, and interpret them. Do the chosen bandwidth and polynomials make sense to you ? why or why not. Third, let's under-smooth. Fix the bandwidth at ½ the size the software gave you, use a linear model on each side (polynomial of 1). Plot them again to see the difference. Do the results differ? How so, if they do? Which do you prefer and why?

The homework first asks us to test for continuity in the covariates. I test this hypothesis using rdrobust and present the results in Table 1. All covariates have large p-values (I test at the p=0.15 level to be consistent with how the homework asks us to test randomization later) as well as minor estimated effects. This implies no significant effect on the covariates, confirming continuity. However, I have also attached figures showing the covariate continuity fitted with a second-degree polynomial (Figures 2 though 6).

Table 1: Covariate Continuity Test Results

Covariate (Outcome)	**RD Estimate ($\hat{\tau}$)**	**p-value**	**Continuous**
ForgnPct	0.0347	0.704	Yes
GovWkPct	0.1055	0.773	Yes
BlackPct	-0.5245	0.433	Yes
VtTotPct	-1.2005	0.172	Yes
GovDem	0.0020	0.898	Yes

Note: $\hat{\tau}$ represents the estimated discontinuity in the covariate at the cutoff ($margin = 0$). All covariates exhibit strong balance, as the p-values are greater than the $\alpha = 0.15$ significance level.

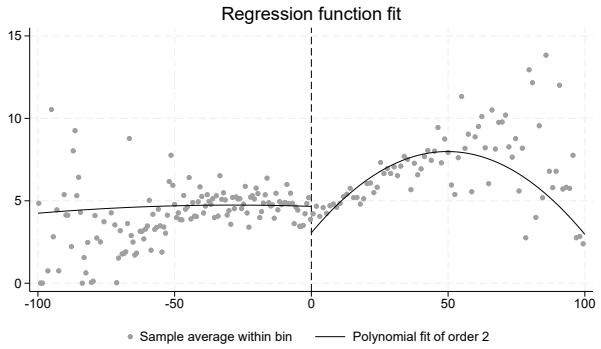

Figure 2: Percent of Population that are Foreigners

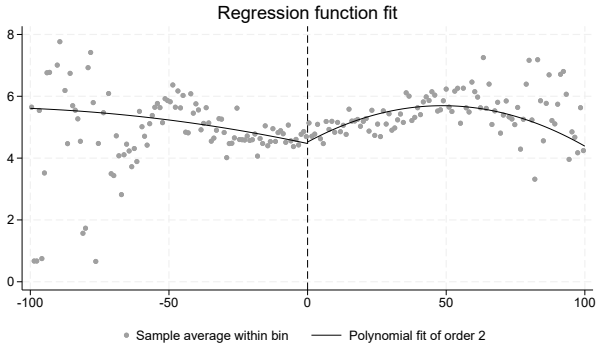

Figure 3: Percent of Population that are Government Workers

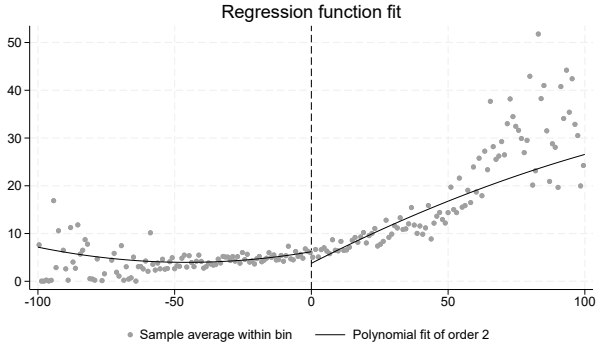

Figure 4: Percent of Population that are Black

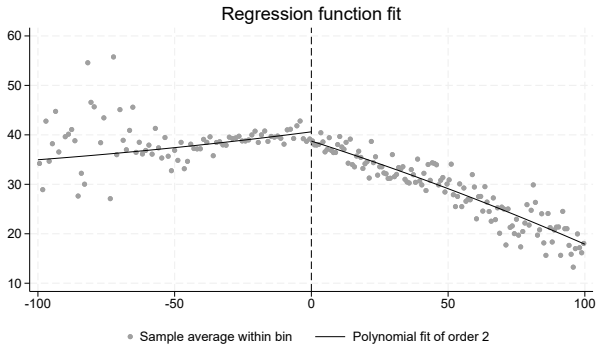

Figure 5: Total Votes as a Percentage of Population

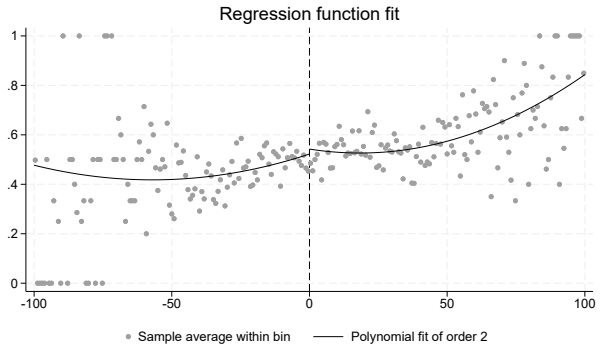

Figure 6: Democratic Governor

Second, the homework asks us to estimate the treatment effect with the optimal bandwidth for a first-degree polynomial fit. We are then asked to double this bandwidth for a second-degree polynomial fit and finally to cut the bandwidth in half for a degree zero polynomial fit. These results are given in Table 2. As the table shows, the "optimal" bandwidth is 26.579, which I then double and then half for the p(2) and p(0) models, respectively. All three have highly significant results, ranging between 15 and 22. This suggests that a Democrat who is an incumbent gets on average between 15 and 22% more of the vote just because of his **incumbent** status. I also Present Figures 7 through 9 which show the discontinuity in result for each specification.

Table 2: Estimated Effect: Incumbency Advantage Sensitivity

Regression Discontinuity Design (RDD)				
Model	**Poly.** (p)	**Bandwidth** (h)	**Estimated Effect** ($\hat{\tau}$)	p**-value**
Optimal Benchmark	1	± 26.579	16.180	< 0.001
Polynomial Sensitivity	2	± 53.158	15.537	< 0.001
Constant Sensitivity	0	± 13.290	22.595	< 0.001

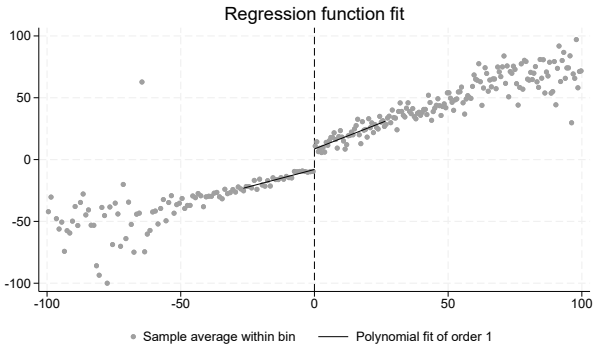

Figure 7: Polynomial of Degree One

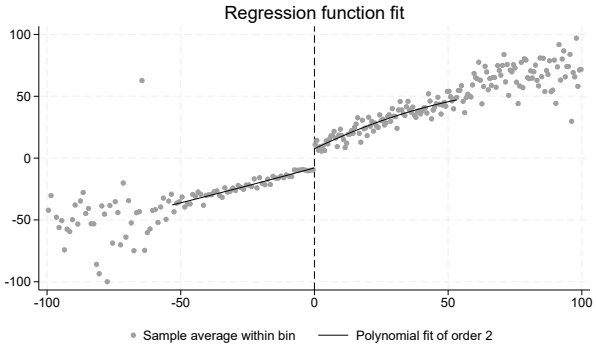

Figure 8: Polynomial of Degree Two

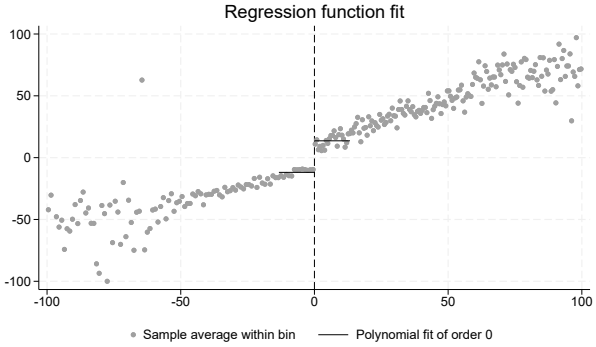

Figure 9: Polynomial of Degree Zero

Finally, for the original p(1) optimal bandwidth model, the homework asks us to run the model again with the bandwidth cut in half. The result is presented in Table 3 and Figure 10. As the table shows, the estimate is still highly significant with an almost identical estimate. I prefer the model with the truncated bandwidth. The reason is because in this project we are – at least in name – looking at "close elections." If we use a bandwidth of 26, that implies that elections that were runner-up 24% to winner 76% are in our analysis "close." This is non-sense. Even the truncated bandwidth of 13 is ridiculous – no one thinks an election decided 37% to 63% is "close" in any way.

Table 3: Bandwidth Sensitivity Check ($p = 1$)

Model	**Poly.** (p)	**Bandwidth** (h)	**Estimated Effect** ($\hat{\tau}$)	p**-value**
$p = 1$ (Reduced h)	1	13.290	16.737	< 0.001

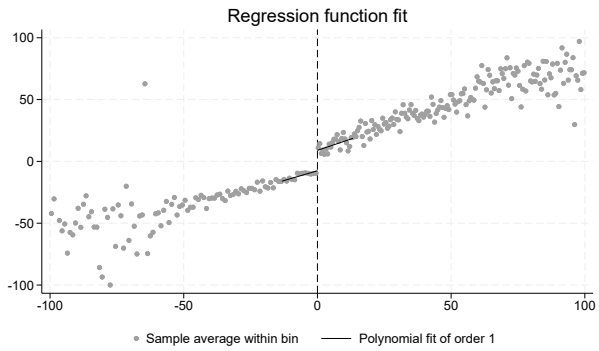

Figure 10: Truncated Bandwidth p(1)

Part 3: Treatment Effect – Local Randomization Assumption

3. Finally, let's use the local randomization approach. Use the 5 given covariates and determine the largest window that fails to reject covariate balance at the 0.15 level. Use rdwinselect. Explain why you are doing this and what it means. Now, estimate the treatment effect and p-value in that chosen window and explain your findings. Use rdrandinf. Finally, cut this window in half and estimate the treatment effect and p-value again with rdrandinf. Discuss any changes.

This part of the homework asks us to test if this model gives us a similar result under the local randomization specification. The homework first asks us to find the largest bandwidth that fails to reject local randomization. To do this I use rdwinselect which automatically selects the window for you. I ran this while also dropping observations with missing covariates ("dropmissing") and also increase the number of windows it tests to find a robust result (I use n=100, default is n=10). The rdwinselect package seems to be conservative and will only recommend the last bandwidth before you first fall below the p=0.15 threshold – the homework, however, asks us to use the **last** bandwidth that meets the criteria. I find that the largest bandwidth that does not fail to reject local randomization is a kernel of ± 3.690. We are then asked to again cut this bandwidth in half and estimate the effect over that truncated kernel. The results are in Table 4. These two estimates show a strong effect, while making the "close election" conjecture believable (elections decided by 7% or less and, in the truncated analysis, 3.5% or less). The estimated effect suggests the margin boost for being an incumbent is around 20% of the vote. The results are robust in significance, and also still have a respectable number of observations.

Table 4: Local Randomization Window Sensitivity

Model	**Bandwidth**	**Total Obs.**	**Estimated Effect ($\hat{\tau}$)**	**p-value**
Wider Window	± 3.690	591	19.513	< 0.001
Narrower Window	± 1.845	295	20.456	< 0.001

Part 4: Personal Preference

4. Which approach, continuity or local randomization do you prefer for studying close elections and explain why?

For close elections, I think local randomization is the superior method. It is much more strict and forces you into a smaller bandwidth (3.690 rather than 26.579!). This is much more believable for "close elections," but beyond this, the continuity assumptions hold assuming a small bandwidth. Given the changes in margin we see in the covariates such as percent black and foreigner there is no reason to believe these candidates are comparable when you move out to a bandwidth that is so large. Local randomization forces you to take seriously the need for covariates to be comparable in a way that continuity does not (specifically when coupled with "optimal" bandwidths). I think both methods could be used well, however, if the user exercises some amount of common sense in overriding absurd bandwidths that are spit out from the models.